

Harikrishna Jiju

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Professional Summary

ML Engineer with 3+ years of experience building and deploying production-grade AI/ML systems at a global financial institution. Specialized in multi-agent orchestration, retrieval-augmented generation (GraphRAG), LLM evaluation pipelines, and cloud-native ML infrastructure serving thousands of enterprise users in production. Proficient in Python, Java, and ML infrastructure across Azure and GCP. Passionate about scalable, responsible AI systems that drive real-world impact.

Education

Deakin University <i>Master of Science in Data Science (Global) AQF Level 9, WES Accredited</i> • Focus: Advanced AI & Scalable ML Systems	Jul. 2025 – Oct. 2027 (Expected) <i>Australia</i>
Indian Institute of Technology (IIT) Madras <i>Diploma in Data Science</i> • Coursework: Mathematics for AI, Statistics, ML Algorithms	May 2024 – Jul. 2025 <i>Online</i>
Vellore Institute of Technology (VIT) <i>B.Tech in Computer Science and Engineering GPA: 9.19/10.0</i>	Jul. 2019 – Jun. 2023 <i>Amaravati, India</i>

Experience

Standard Chartered Bank <i>Machine Learning Engineer – GenAI</i> • Built an ML infrastructure evaluation and experiment-tracking framework on MLflow, instrumenting application pipelines so prompt/config changes auto-register as tagged experiments with continuous metric monitoring. • Led early-stage fine-tuning of small open-source LLMs on on-premise NVIDIA GPU infrastructure, evaluating parameter-efficient tuning approaches for domain adaptation and model optimization. • Architected and led production deployment of a multi-agent Regulatory-Change-to-Policy (R2P) AI system using LangGraph, integrating a hybrid GraphRAG architecture (Nebula Graph + Azure AI Search) that reduced manual policy impact assessment time by 90% bank-wide. • Engineered async processing pipelines using Python <code>asyncio</code> and <code>ThreadPoolExecutor</code> across multi-stage LLM workflows, reducing end-to-end agent response latency by 80% in production. • Owned full SDLC for the R2P agent: system design using Clean Architecture (Django REST Framework), SIT, UAT, performance testing, and production release deployed to all employees globally. • Built and deployed an enterprise conversational agent (LangGraph multi-agent orchestration) with streaming SSE responses, persistent memory, guardrails (Azure Content Safety), and Azure Databricks vector search, achieving <3s P95 response time at scale. • Developed LLMOps evaluation pipelines using Ragas (context precision, answer relevancy, faithfulness) and custom mathematical metrics to benchmark GPT-4, GPT-4o, GPT-5, GPT-5.1, Claude 3.5 Sonnet, Claude 4.6 Sonnet, and Claude 5 models on financial Q&A tasks. • Implemented structured prompt engineering (chain-of-thought, tool-use, few-shot) and input sanitization layers for PII detection and content filtering in production LLM applications. • Collaborated with compliance, risk, and technology stakeholders through design reviews to align AI solutions with enterprise standards and responsible AI principles.	Jul. 2023 – Present <i>Bengaluru, India</i>
Kimberly-Clark <i>Software Engineer Intern</i> • Designed and implemented an NLP + OCR invoice parsing solution (Tesseract) that automated document processing for the South African market, improving throughput by 80%.	Jan. 2023 – Jun. 2023 <i>Bengaluru, India</i>

Projects

LLM Evaluation Harness <i>Python, MLflow, Ragas, Ollama, LangChain, FastAPI, SQLite</i> • Built a modular RAG evaluation framework using open-source LLMs via Ollama (Llama 3, Mistral) and Ragas metrics (context precision, answer relevancy, faithfulness) benchmarked on public QA datasets. • Exposed evaluation results via a FastAPI service with SQLite for run history and metric comparison across prompt/model variants, enabling reproducible regression testing.	
Scalable Document Q&A Pipeline <i>Python, LangChain, FAISS, GCP Cloud Run, Vertex AI, BigQuery</i> • Designed an end-to-end document ingestion and retrieval pipeline: PDF chunking, embedding via Vertex AI (<code>text-embedding-004</code>), FAISS vector index, and a LangChain RetrievalQA chain for semantic Q&A. • Deployed inference API on GCP Cloud Run (serverless, auto-scaling); benchmarked retrieval quality across chunk sizes and models, achieving 88%+ answer relevancy on a 500-document corpus.	

Technical Skills

Languages: Python, Java, SQL, C++
ML & AI: LLM Fine-tuning, Multi-Agent Orchestration (LangGraph), GraphRAG, RAG, Prompt Engineering, NLP, Ragas Evaluation
ML Infrastructure & LLMOps: Model Deployment, MLflow, Azure AI Search, Azure Databricks, OpenAI (GPT-4/4o/5/5.1), Anthropic (Claude 3.5/4.6/5), Vector DBs (FAISS, Pinecone, Weaviate), Nebula Graph
Cloud & DevOps: GCP (Vertex AI, BigQuery, Cloud Run), Azure, Docker, Kubernetes, CI/CD, Azure DevOps
Frameworks: Django REST Framework, FastAPI, SpringBoot, LangChain